

G Pavan Kalyan | Hardware Engineer

Bengaluru, Karnataka – India

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Professional Summary

Results-driven Hardware Engineer with 2+ years of experience in automotive electronics and EV systems. Specialized in Body Control Module (BCM) design, DC-DC converters, and Battery Management Systems (BMS). Proven expertise in hardware-software integration, cost optimization, and regulatory compliance. Proficient in advanced design tools including ORCAD, Cadence, and Vector CAN analysis tools.

Professional Experience

Simple Energy Pvt Ltd

Hardware Engineer

Bengaluru, India

Sep 2023–Present

Key Achievements:

- **BCM Development:** Designed and implemented Body Control Module hardware solutions for automotive applications, managing electronic accessories and vehicle functions
- **DC-DC Converter Design:** Leading hardware development of DC-DC converters for EV two-wheelers, ensuring optimal power conversion efficiency
- **Cost Optimization:** Drove BMS cost reduction initiatives, achieving significant savings while maintaining performance standards
- **Testing & Validation:** Developed and executed comprehensive test plans for DC-DC converters, ensuring reliability and compliance
- **Technical Analysis:** Performed advanced CAN communication protocol analysis using Vector tools for network optimization

Technical Contributions:

- Collaborated with software teams to ensure seamless hardware-software compatibility
- Conducted root cause analysis for complex circuit board and network configuration issues
- Generated detailed technical reports on test results, design modifications, and product specifications
- Recommended cost-effective hardware solutions aligned with customer requirements
- Managed complete product lifecycle from schematic capture to prototype development

Technic Control Facility Management System

Graduate Engineer Trainee

Bengaluru, India

May 2023–Jul 2023

- Specialized in transformer and substation systems design, operation, and maintenance protocols
- Authored comprehensive customer reports with technical insights and actionable recommendations
- Managed Low Tension (LT) and High Tension (HT) electrical systems operations
- Conducted electrical component audits ensuring safety compliance and operational efficiency

JSW Steels

Graduate Engineer Trainee

Toranagallu, India

Mar 2020–Nov 2020

- Monitored and optimized manufacturing equipment performance in steel production environment
- Conducted site surveys, precision measurements, and comprehensive project condition assessments
- Performed quality inspections of completed construction projects ensuring specification compliance

Key Projects

Academic Project

Wireless Power Transfer System for Electric Vehicles

2023

Project Overview: Designed innovative wireless charging system for EVs using inductive power transfer technology

Technical Implementation:

- Developed AC-DC converter and high-frequency inverter systems
- Engineered transmitting coil (road-embedded) and receiving coil (vehicle-mounted) coupling mechanism
- Implemented dynamic charging capability for vehicles in motion through magnetic field induction
- Achieved efficient power transfer for battery charging applications

Technologies Used: Power Electronics, Electromagnetic Design, Battery Management Systems

Education

Reva University

Bachelor of Technology (B.Tech), Electrical & Electronics Engineering

Bengaluru

2020–2023

Sir M Visvesvaraya Polytechnic

Diploma, Electrical & Electronics Engineering

Hospete

2016–2019

Technical Skills

Software & Tools.....

- ORCAD & Cadence (Circuit Design & Simulation)
- Vector CANoe/CANalyzer (Network Analysis)
- MATLAB/Simulink (System Modeling)
- AutoCAD (Technical Drawings)

Hardware Expertise.....

- Microcontrollers & Embedded Systems
- Power MOSFETs & Power Electronics
- PCB Design & Layout
- Electronic Component Selection & Testing

Communication Protocols.....

- CAN (Controller Area Network), LIN (Local Interconnect Network), I2C, SPI, UART

Core Competencies.....

- Automotive Electronics (BCM, BMS, DC-DC Converters)
- Circuit Design & Hardware Architecture
- System Integration & Testing
- Root Cause Analysis & Troubleshooting
- Cost Optimization & Component Selection
- Cross-functional Collaboration

Languages

English: Professional Proficiency

Hindi: Native/Bilingual

Kannada: Native/Bilingual

Telugu: Native/Bilingual